

introduced a course on 'Drone Piloting and Technician' in July 2020 (when corona lockdowns were being done away with).

The last few years, as per Singh, saw people trying to push boundaries of drones, keeping the service industry in consideration. A lot of work has been done to make drones more intelligent and agile. Such work has been on the dominant side in the last couple of years in India. A lot in terms of data collection and churning has also been the focus during the last few years, as explained by Singh.

"People are now developing more sector-focused applications for drones. The future of drones is all about how drones can improve the profitability in business and increase productivity at the same time, and that is going to come when we extract more information from the data. Everything happening around drones at the moment is business-centric," adds Singh.

A study by Fortune Business Insights has projected the global drone services market size to reach \$67.43 billion by 2026, exhibiting a CAGR of 41.34 per cent during the forecast period (2020 to 2026). This market size was valued at \$4,233.9 million in 2018.

The study points out that the utilisation of drones in military applications will boost this market revenue. "The rising demand for commercial drones in agricultural applications such as crop rotation, crop protection aquaponics, and organic farming will accelerate the drone services market growth. Besides, the increasing requirement for data analytics, data acquisition, and real-time inspection will spur demand for drones, which, in turn, will enable speedy growth of the market in the forthcoming years," reads the study.

"There are new verticals coming up in the drone services industry. These range from higher payload capable drones to long-range delivery of goods using drones to aerial taxis, and so on. I think standardisation is what will take place in the next couple of years," mentions Singh.

Giving an example of land surveying by drones, Singh notes that while the likes of him were trying to prove to the government that land surveying through drones is possible, the government now has introduced a pro-

Speakers

Anurag Joshi,
co-founder,
INDrone Aero
Systems



A.T. Kishore,
CEO,
Vidya Sangha
Technologies

John Livingstone,
founder CEO &
product architect,
Johnnette
Technologies Pvt Ltd



Mughilan T.
Ramasamy,
co-founder,
Skylark Drones

Tanuj Bhojwani,
ispirt



Varun Bhatia,
Head - Standards,
Electronics Skill
Sector Council of
India (ESSCI)

Vipul Singh,
founder and CEO,
Aarav Unmanned
Systems



gramme that includes use of drones for doing land survey of villages in India. The number of villages in India is approximately 660,000. Singh compares the ecosystem of drones to the ecosystem of smartphones. He notes that just like the smartphone ecosystem has developed itself into

a big business opportunity, the same will happen to the drone's ecosystem. The opportunities will include manufacturing, hardware, battery, software, sensors, and even apps designed around drones.

Land survey using drones is just the tip of the iceberg as per Joshi. He is of the view that once beyond line of sight (BLS) flights for drones are allowed, the business opportunities will increase at a higher pace than what India is witnessing now. These opportunities, as explained by Joshi, would include disaster management, surveying pipelines, solar farms, logistics, and more. These operations usually include sending drones 50 to 60 kilometres away from the pilot location.

"A lot of industry sectors know that there is a need, but they are still hesitant to adopt drones as the understanding of the technology is not there. However, if we look at the evolution, it has been phenomenal. From using drones as flying camera devices to now engineering these for focused tasks, the industry has evolved very quickly. Even the number of people consuming data collected by drones has become better," as per Joshi.

He adds, "The amount of selling required to make people understand the benefits of drones is increasing as a lot of learning from the use-cases is taking place."

The current value of different parts of the drone ecosystem, as per a global study by Price Waterhouse notes that the drone infrastructure is valued at \$45.2 billion, drones in transport services is valued at \$13 billion, drones in media and entertainment services is valued at \$8.8 billion, drones in telecommunication services is valued at \$6.3 billion, drones in agriculture services is valued at \$32.4 billion, drones in security services is valued at \$10.5 billion, drones in insurance services is valued at \$6.8 billion, and drones in mining services is valued at \$4.3 billion. **EFY**

This article is based on a panel discussion around drones that took place during the December 2020 edition of India Technology Week. The discussion was moderated by Neel Mehta, director & co-founder, Asteria Aerospace, and Vichar Shroff, co-founder, burncal.a